ECO-HABITAT HEALTH TRACKER

A MINI-PROJECT REPORT

Submitted by

KARTHAVYA S 240701229

ILAKKIYA P 240701194

in partial fulfillment of the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institute

CHENNAI

OCTOBER 2025

BONAFIDE CERTIFICATE

Certified that this project“”is **ECO-HABITAT HEALTH TRACKER** the bonafide work of **“KARTHAVYA S, ILAKKIYA P”** who carried out the project work under my supervision.

SIGNATURE

Dr. V. JANANEE

ASSISTANT PROFESSOR SG

Dept. of Computer Science and Engineering,
Rajalakshmi Engineering College
Chennai

SIGNATURE

Dr. Deepa Sai

ASSISTANT PROFESSOR SG

Dept. of Computer Science and Engineering,
Rajalakshmi Engineering College
Chennai

This mini project report is submitted for the viva voce examination to be held on

____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Habitat Tracker is a Java Swing application that monitors wildlife reserves by tracking environmental factors and evaluating risk levels. Users can enter habitat data like temperature, humidity, and pollution readings manually. The system saves this information with timestamps for easy record-keeping. It also provides graphical displays of temperature trends, allowing users to see changes over time. A detailed habitat list and risk-level display help spot potential threats and prompt timely responses. With its interactive interface and data-based insights, Habitat Tracker aids in effective monitoring and proactive conservation of wildlife environments.

ACKNOWLEDGEMENT

We express our sincere thanks to our beloved and honorable chairman **MR. S. MEGANATHAN** and the chairperson **Dr. (Mrs.) THANGAM MEGANATHAN** for their timely support and encouragement.

We are greatly indebted to our respected and honorable principal **Dr. S.N. MURUGESAN** for his able support and guidance.

No words of gratitude will suffice for the unquestioning support extended to us by our Head Of The Department **Dr. E.M. MALATHY** and our Deputy Head Of The Department **Dr. J. MANORANJINI** for being ever supporting force during our project work

We also extend our sincere and hearty thanks to our internal guide **Dr. V. JANANEE ,** for her valuable guidance and motivation during the completion of this project.

Our sincere thanks to our family members, friends and other staff members of computer science engineering.

1. KARTHAVYA S

2. ILAKKIYA P

TABLE OF CONTENTS

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The introduction page presents an overview of the Habitat Tracker system. It highlights its purpose—monitoring wildlife reserves through parameter readings, visual graphs, and risk assessment. Users can easily navigate to view habitats, record data,

and track environmental changes in real time.

1.2 SCOPE OF THE WORK

The Habitat Tracker project focuses on monitoring wildlife habitats by recording environmental parameters, displaying habitat lists, and assessing risk levels. It enables manual data entry with timestamped readings and graphical analysis. The system supports

efficient conservation management and can be expanded for real-time sensor-based monitoring.

1.3 PROBLEM STATEMENT

Wildlife habitats face increasing threats from climate changes, pollution, and human interference, making constant monitoring essential. Traditional methods of tracking environmental conditions are time-consuming and inefficient. There is a need for a digital system that records, analyzes, and alerts risk levels to ensure effective

habitat protection and management.

1.4 AIM AND OBJECTIVES OF THE PROJECT

To develop a Java Swing-based application that monitors wildlife habitats by tracking environmental parameters, assessing risk levels, and supporting proactive conservation management.

Objectives:

- To record and store environmental readings with timestamps.
- To display habitat lists and their respective risk levels.
- To visualize temperature data through graphs.
- To enable manual data entry and secure database management.
- To generate alerts for abnormal or risky habitat conditions

CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	:	Intel i5
Memory Size	:	16GB (Minimum)

HDD	:	64GB (Minimum)

2.2 SOFTWARE SPECIFICATIONS

Operating System	:	WINDOWS 11
Front – End	:	Java Swing
Back - End	:	MySql
Language	:	Java,SQL

CHAPTER 3

MODULE DESCRIPTION

This application consists of a single module. When the program runs, it will display a login window asking for confirmation. The person who interacts can log in as an **Administrator**. The description of the module is
as follows:

Admin Login When the person logs in as an Administrator, they must enter a valid username and password. The administrator has full control over the system, including adding, updating, and deleting habitat details. They can manually enter environmental readings such as temperature, humidity, and pollution levels, view risk levels, analyze graphical trends, and manage the database efficiently.

CHAPTER 4

SAMPLE CODING 1

This section represents the Main Java class that serves as the entry point of the Habitat Tracker application. It initializes the system, launches the login window, and connects the interface with the backend modules such as database handling and habitat monitoring functions.

Main.java

```
import javax.swing.*;

public class Main {

    public static void main(String[] args) {

        SwingUtilities.invokeLater(() -> {

            LoginDialog loginDlg = new LoginDialog(null);

            loginDlg.setVisible(true);

            if (loginDlg.isSucceeded()) {

                new HabitatQualityUI(); // Launch main JSwing UI

            } else {

                System.exit(0); // Exit if login fails or canceled

            }

        });
```

```
    }
}
```

SAMPLE CODING 2

This section represents the Login Page module, which handles user authentication for the Habitat Tracker application. It verifies the administrator's username and password, ensuring secure access to the system. Upon successful login, the main dashboard is launched; otherwise, an error message is displayed prompting re-entry of valid credentials.

Login.java

```
package src.view;

import javax.swing.*;

import java.awt.*;

public class LoginDialog extends JDialog {

    private boolean succeeded;

    public LoginDialog(Frame parent) {

        super(parent, "Administrator Login", true);

        // Main panel with padding

        JPanel panel = new JPanel(new GridBagLayout());

        panel.setBorder(BorderFactory.createEmptyBorder(20, 20, 20, 20));

        GridBagConstraints gbc = new GridBagConstraints();

        gbc.insets = new Insets(10, 10, 10, 10);
```

```
gbc.fill = GridBagConstraints.HORIZONTAL;

// Username

gbc.gridx = 0;

gbc.gridy = 0;

panel.add(new JLabel("Username:"), gbc);

gbc.gridx = 1;

JTextField tfUsername = new JTextField(20);

panel.add(tfUsername, gbc);

// Password

gbc.gridx = 0;

gbc.gridy = 1;

panel.add(new JLabel("Password:"), gbc);

gbc.gridx = 1;

JPasswordField pfPassword = new JPasswordField(20);

panel.add(pfPassword, gbc);

// Buttons

gbc.gridx = 0;

gbc.gridy = 2;

JButton btnCancel = new JButton("Cancel");
```

```
panel.add(btnCancel, gbc);

gbc.gridx = 1;

JButton btnLogin = new JButton("Login");

panel.add(btnLogin, gbc);

getContentPane().add(panel, BorderLayout.CENTER);

// Button actions

btnLogin.addActionListener(e -> {

  String username = tfUsername.getText().trim();

  String password = new String(pfPassword.getPassword()).trim();

  if ("admin".equals(username) && "admin123".equals(password)) {

    succeeded = true;

    dispose();

  } else {

    JOptionPane.showMessageDialog(LoginDialog.this,

        "Invalid username or password",

        "Login Failed",

        JOptionPane.ERROR_MESSAGE);

    tfUsername.setText("");

    pfPassword.setText("");
```

```
    succeeded = false;

} });
```

CHAPTER 5

SCREEN SHOTS

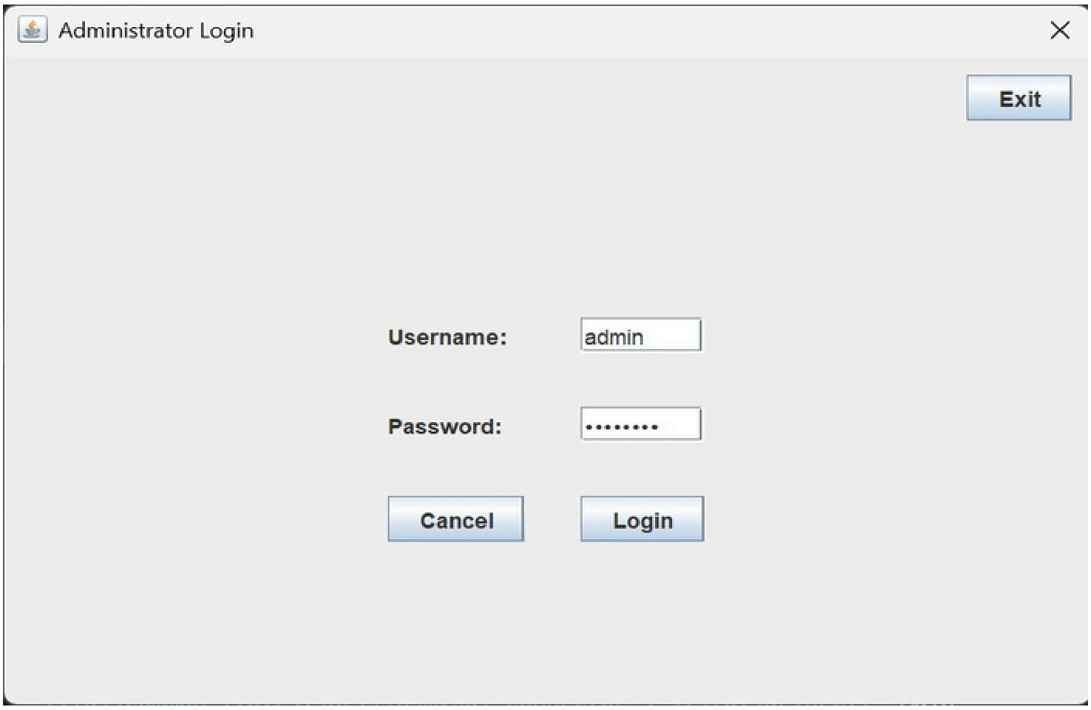

Fig 5.1 Introduction page

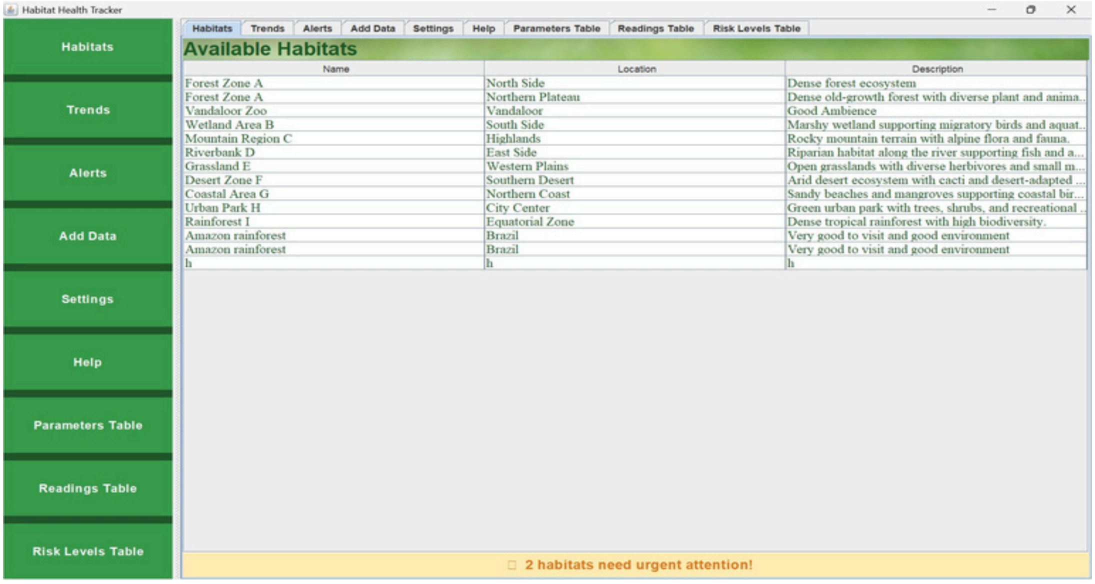

Fig 5.2 Habitats list

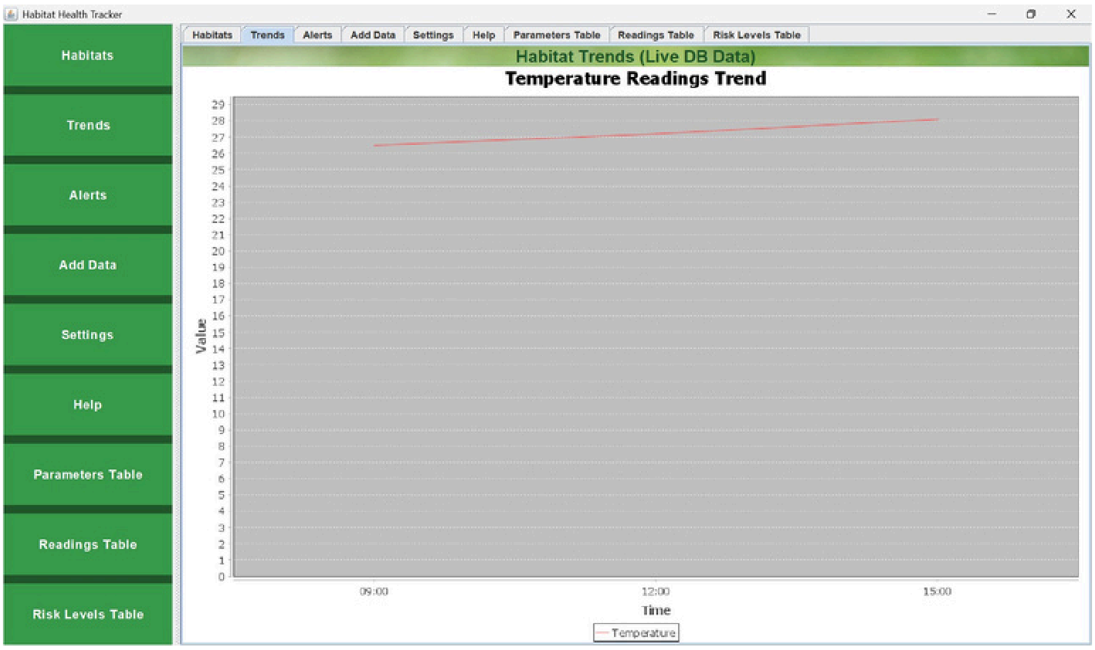

Fig 5.3 Trend Analysis

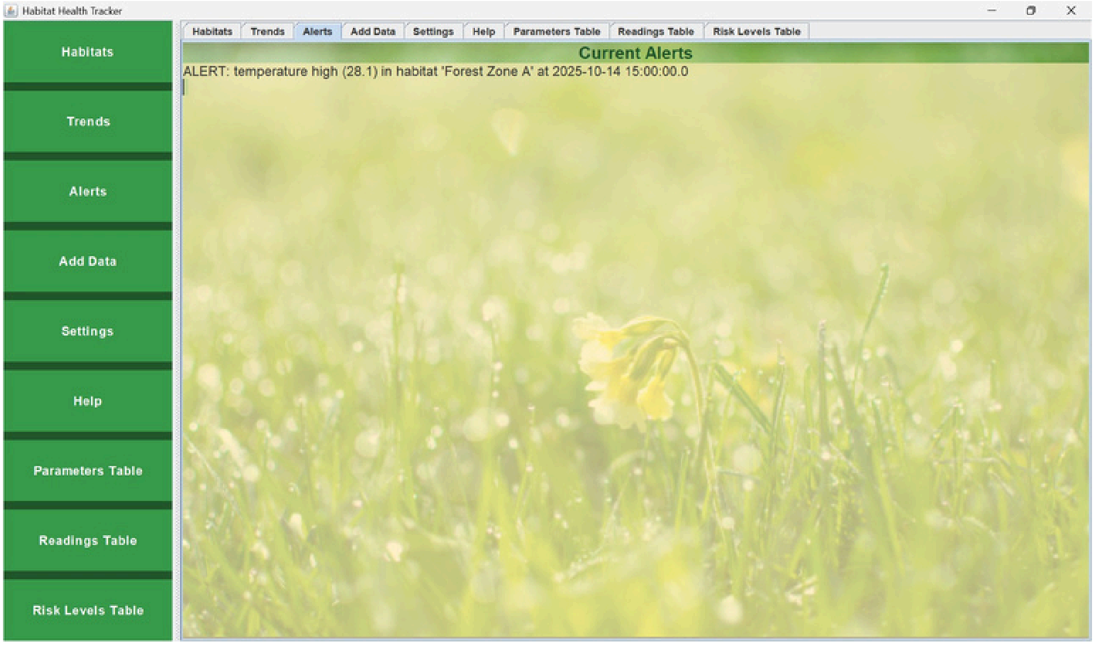

Fig 5.4 Alerting System

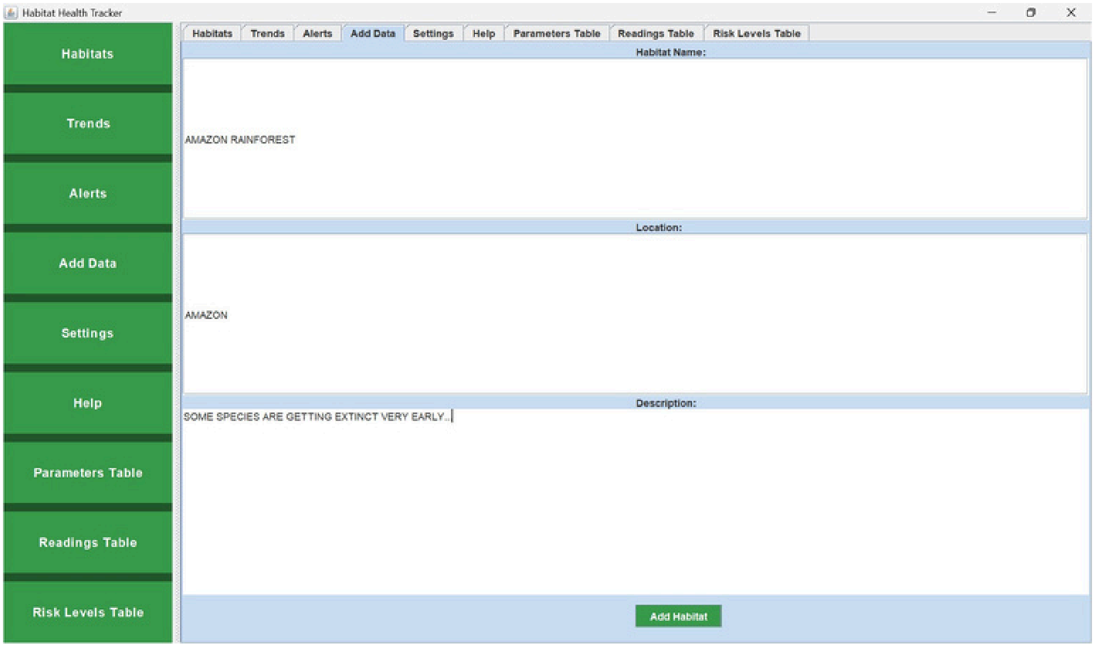

Fig 5.5 Data Addition

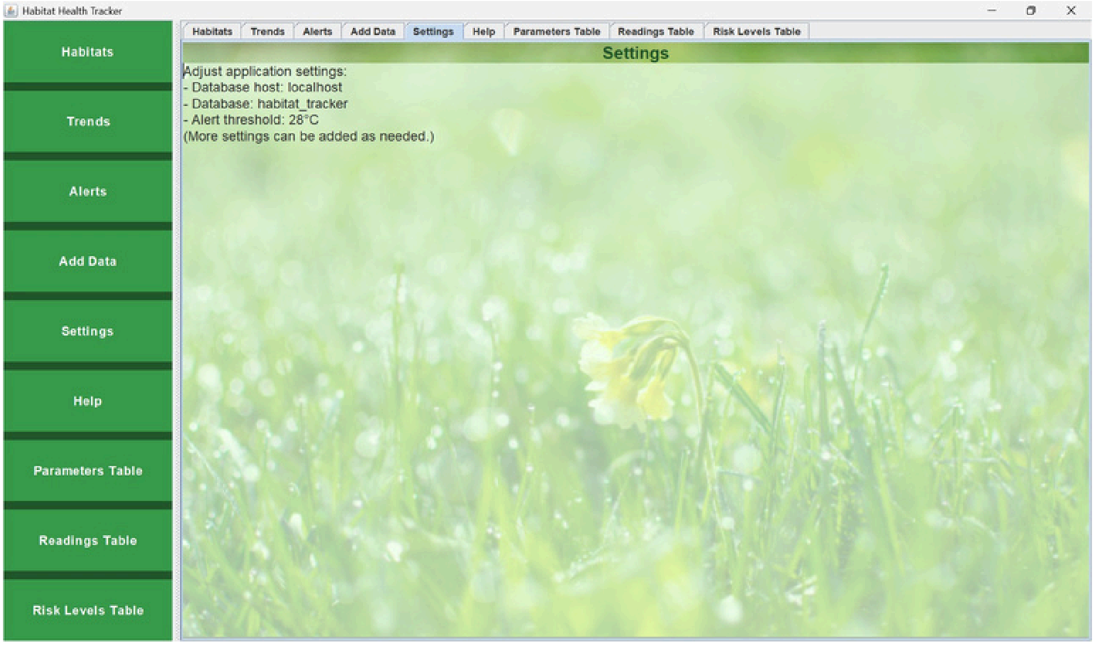

Fig 5.6 Settings

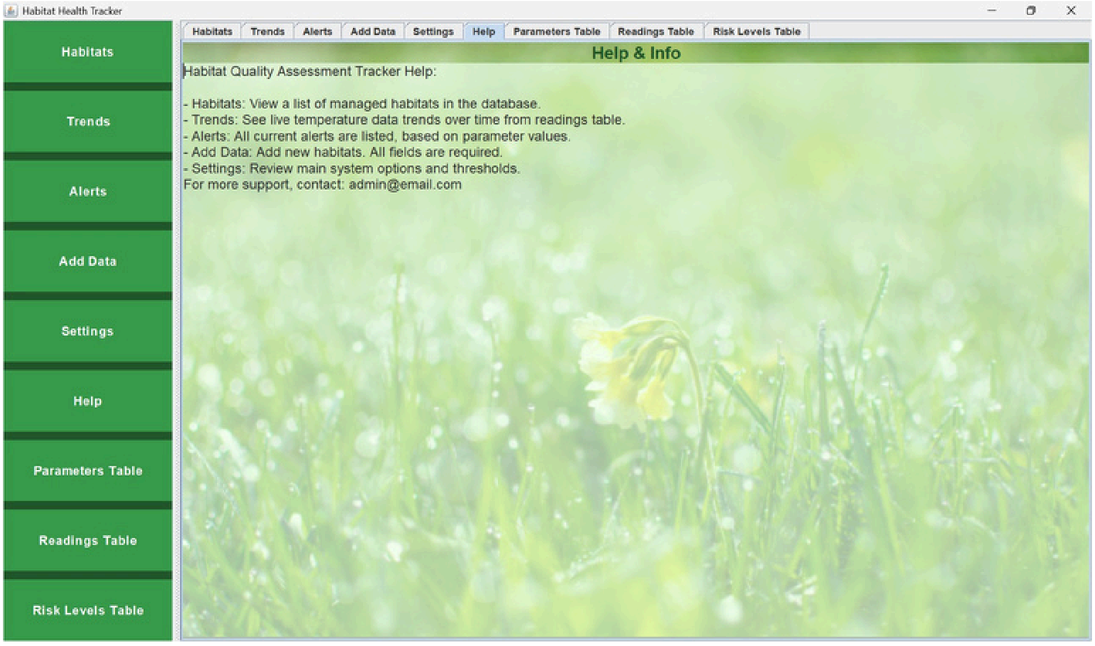

Fig 5.7 Help Tab

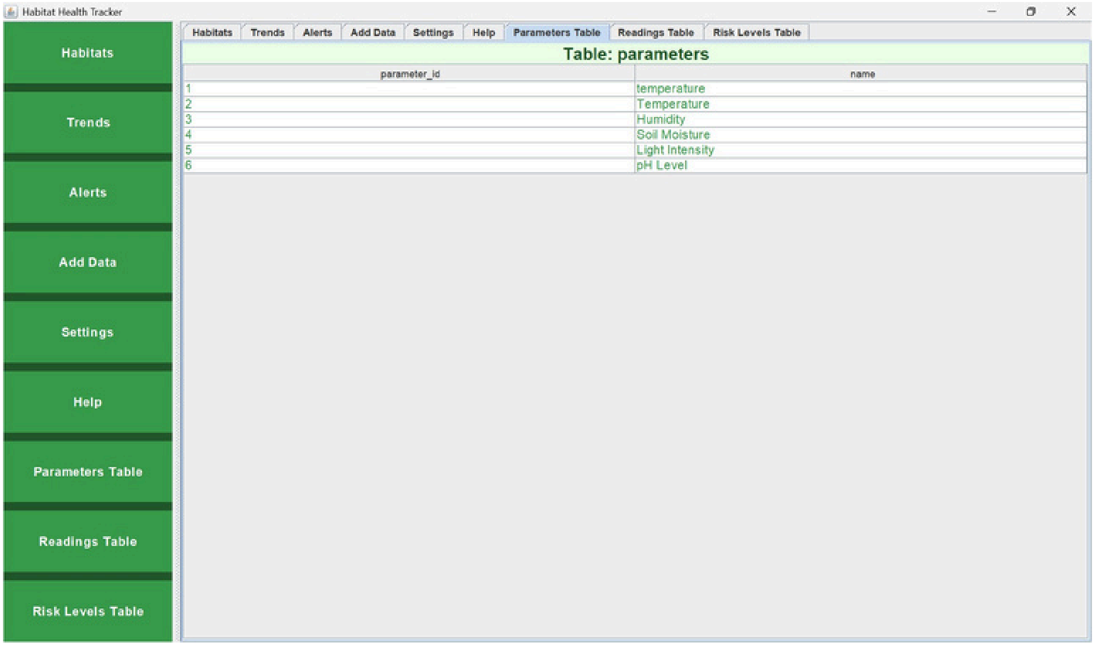

Fig 5.8 Parameters Table

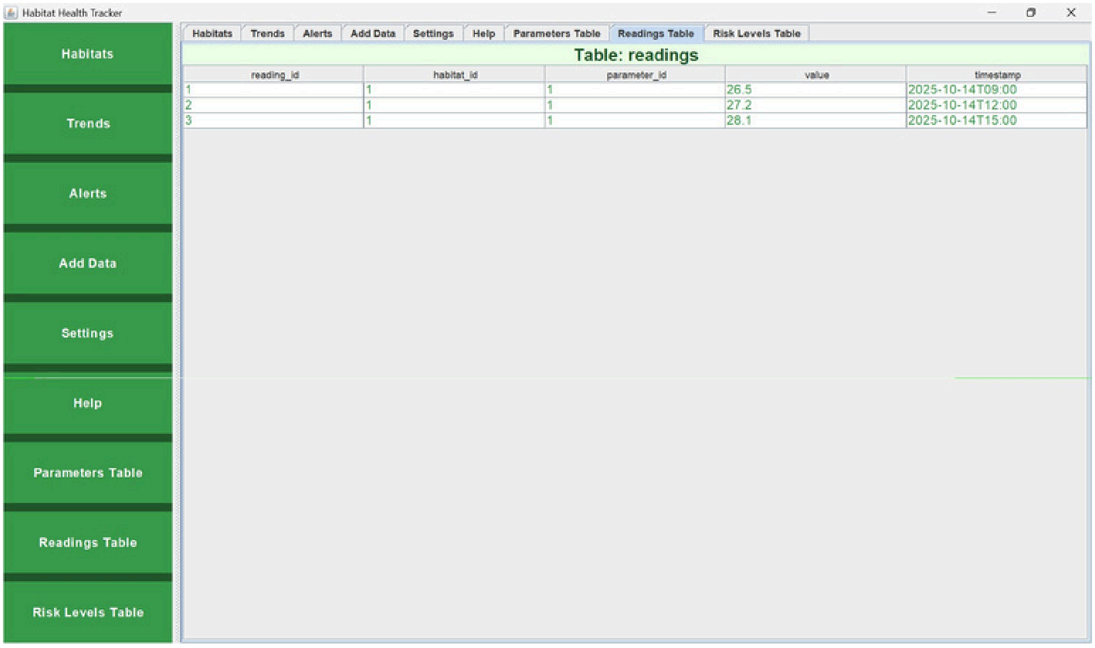

Fig 5.9 Readings Table

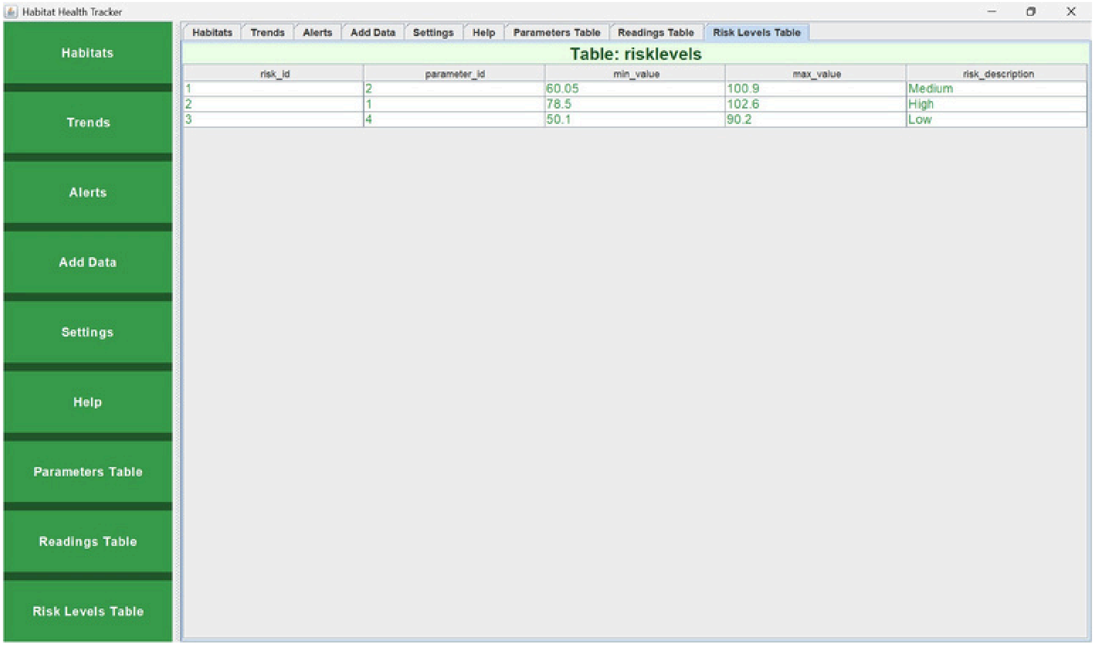

Fig 5.10 Risk Levels Table

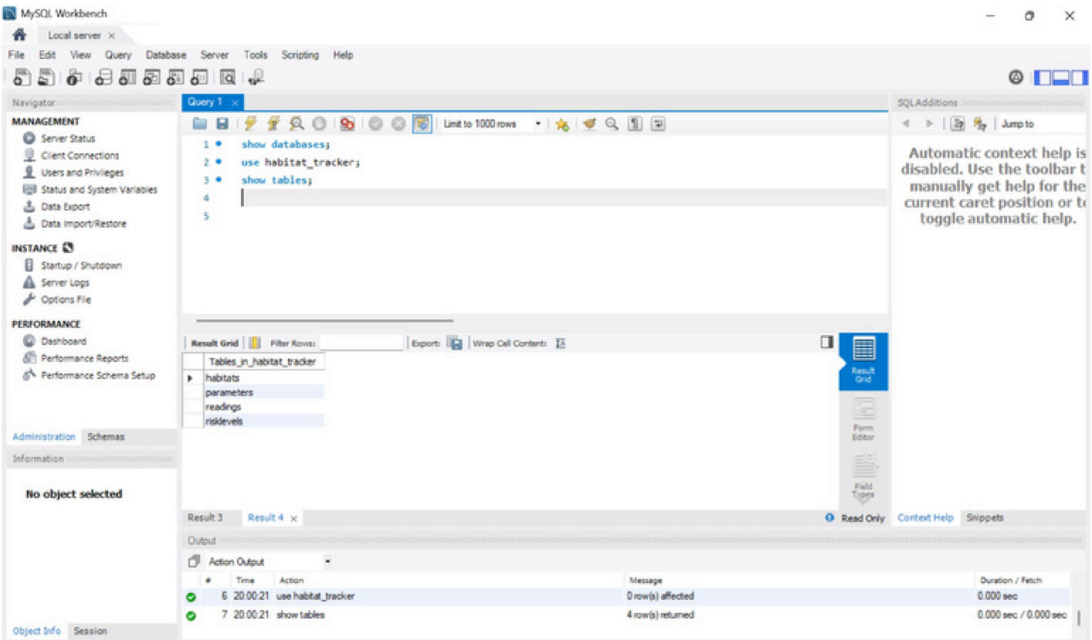

Fig 5.11 Database Creation(Tables)

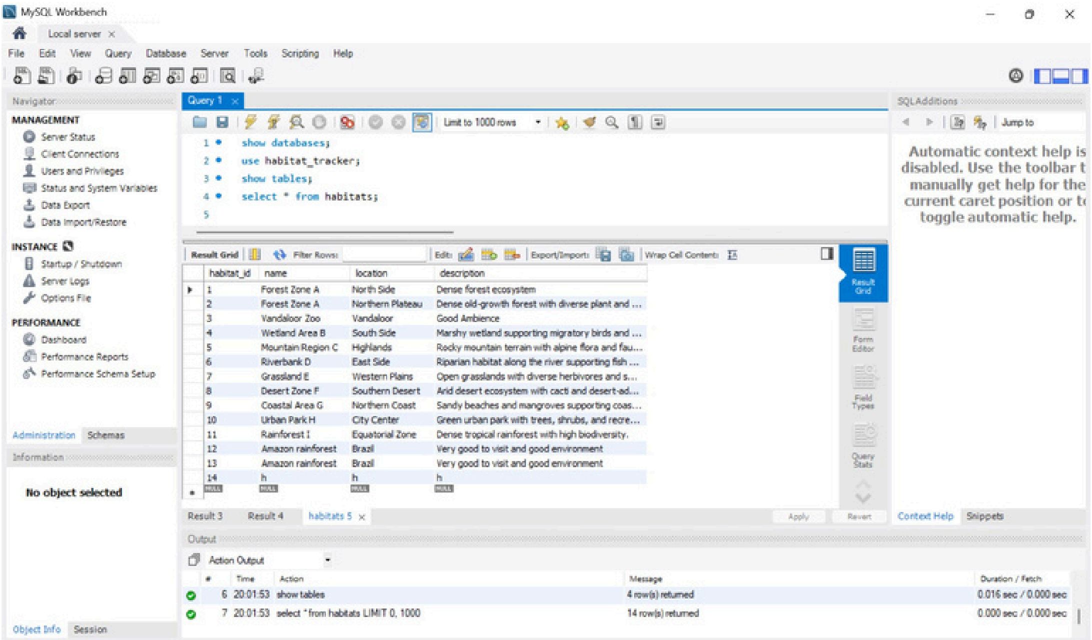

Fig 5.12 Database Creations(Values)

CHAPTER 6

CONCLUSION AND ENHANCEMENTS

The Habitat Tracker project successfully demonstrates how technology can aid wildlife conservation by monitoring environmental parameters and assessing habitat risk levels. The system provides an easy-to-use interface for administrators to record, view, and analyze data effectively. It ensures timely awareness of changes through readings and visual graphs. Future enhancements can include integrating IoT sensors for automatic data collection, real-time cloud synchronization, and *mobile application support. Additional AI-based analysis could also be implemented to predict potential threats and provide preventive alerts, making the system more intelligent and impactful for large-scale habitat management.*

REFERENCES

1. https://www.w3schools.com/sql/
2. http://www.extinctblog.org/extinct/tag/habitat+tracking
3. https://www.geeksforgeeks.org/java/introduction-to-java-swing/
4. https://docs.oracle.com/javase/tutorial/uiswing/
5. https://news.umich.edu/how-animal-tracking-data-can-help-preserve-biodiversity/